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RESEARCH ARTICLE

SAM Translator: Self-Paced Learning And Mixture-of-Experts for Cross-Lingual Text-to-Speech Translation

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ABSTRACT Most existing translation studies have concentrated on text-to-text or speech-to-text tasks, whereas text-to-speech translation (T2ST) remains relatively underexplored, despite its central role in cultural integration and education. Among the few available T2ST systems, recent end-to-end approaches still face significant challenges in modeling the linguistic and acoustic diversity that exists in both high-resource and low-resource languages. To address these challenges, we propose SAM-Translator, a novel framework based on the mBART model that incorporates two key mechanisms: Self-Paced Learning (SPL) And Mixture-of-Experts (MoE). SPL enhances training efficiency and stability by progressively introducing samples, from easy to difficult ones, based on training loss. Meanwhile, the MoE mechanism allows the model to dynamically select specialized experts for prediction, thus improving adaptability to cross-lingual linguistic variation. In addition, we tackle the low-resource language problem using the Data-Augmented Self-Training (DAST) approach, which generates synthetic parallel data from monolingual corpora to boost performance in data-scarce scenarios. Experimental results on the CVSS-C benchmark demonstrate that SAM-Translator achieves state-of-the-art performance, yielding relative BLEU score improvements of 2–3% for high-resource and 7–10% for low-resource languages. Remarkably, DAST achieves significant gains for Welsh and Mongolian, with relative BLEU improvements of 21.55% and 25.83%, respectively. Furthermore, evaluations on the large-scale CoVoST-2 dataset confirm the model's robustness, with an additional 4–6% improvement over the baseline. These results underscore the effectiveness of SAM-Translator in advancing multilingual T2ST.

INDEX TERMS Text-to-speech translation (T2ST), self-paced learning (SPL), data-augmented self-training (DAST), mixture of experts (MoE), cross-lingual speech synthesis, acoustic unit discovery.

I. INTRODUCTION

Text-to-Speech Translation (T2ST) converts written text in a source language into spoken speech in a target language, serving as a crucial technology for cross-lingual communication in a globally interconnected world. T2ST has diverse applications, particularly in cultural integration and education. For instance, a traveler in Japan can scan a historical plaque written in Japanese and hear its

description in their native language, providing a deeper cultural experience without requiring proficiency in the source language. Similarly, in countries such as India, China, and Vietnam, where many ethnic minority languages coexist with a majority or official language, T2ST plays a vital role in bridging communication gaps. Specifically, many minority languages lack a standardized written form, preventing the use of conventional text-based translation methods. In such contexts, T2ST can support the dissemination of public information and the expansion of access to education, which promotes fairness and equity.

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A common approach of T2ST is to use a cascaded pipeline, where source-language text is first translated into the target language using Neural Machine Translation (NMT) and the followed by speech synthesis via a Text-to-Speech (TTS) system. This pipeline, exemplified by systems such as Google Translate, relies on intermediate text representations. Recent research has explored end-to-end T2ST models that bypass intermediate text by leveraging speech unit tokens. This design is particularly beneficial for low-resource languages and those lacking a standardized writing system. For example, Mingote et al. [1] proposed a direct T2ST system that translates source-language text into target-language speech without generating intermediate text. Their method tokenizes target speech into discrete unit sequences using an acoustic unit discovery model such as HuBERT [2], and trains a Text-to-Unit (T2U) model based on mBART [3], [4] to map the source text to these unit sequences. A subsequent Unit-to-Speech (U2S) synthesizer then reconstructs the final speech waveform, as illustrated in Figure 1.

Despite promising results, the existing approach still faces performance limitations due to the challenges of learning linguistic and acoustic diversity across languages. To address this, we propose SAM-Translator, a new framework incorporating two key training mechanisms: Self-Paced Learning (SPL) [5], [6] and Mixture-of-Experts (MoE) [7], [8] for the T2ST task. SPL improves cross-lingual learning by gradually adding training samples from easy to hard, based on model loss, which facilitates more stable and effective convergence. MoE enhances model flexibility by dynamically routing each input to a subset of specialized expert modules, allowing better adaptation to the syntactic, semantic, and prosodic characteristics of diverse languages. However, the model's performance in low-resource language settings remains limited by the scarcity of supervised parallel data. To overcome this limitation, we employ a Data-Augmented Self-Training (DAST) method to generate synthetic parallel corpora from monolingual data, thereby enhancing performance, particularly in low-resource language scenarios.

Our contributions can be summarized as follows:

- We propose the novel SAM-Translator framework to achieve state-of-the-art performance for T2ST on the CVSS-C and CoVoST-2 dataset.
- By incorporating DAST, our proposed framework, SAM-Translator, achieves significant performance gains for low-resource languages.

A demonstration and code of this approach is publicly available at https://github.com/tranquangchung/sam_translator.

II. RELATED WORK

A. TEXT-TO-SPEECH

In recent years, deep learning has significantly advanced Text-to-Speech (TTS) systems, especially in improving the naturalness and intelligibility of synthesized speech. End-to-end neural models [9], [10] have outperformed

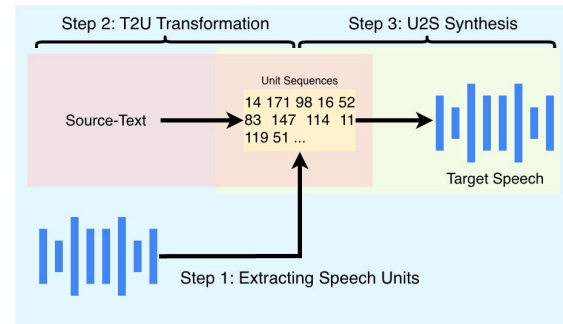


FIGURE 1. Process of converting source text into target speech. In this research, we use text from N languages in CVSS-C and CoVoST-2 to generate English speech.

traditional approaches such as concatenative synthesis [11] and statistical parametric methods [12], [13] in efficiency, generalization, and quality. Pioneering models such as Tacotron2 [14], FastSpeech2 [9], and VITS [10] have marked substantial breakthroughs in the quality of synthesized speech. Alongside these developments, research has progressed in advanced directions such as adaptive TTS [15], low-resource TTS [16], and expressive TTS [17]. While many systems support single or multiple languages, most related work remains focused on intra-lingual TTS, where both the input text and the generated speech are in the same language. This limits their applicability to cross-lingual scenarios. Therefore, in this research, we propose a novel model, SAM Translator, that extends the capabilities of TTS systems for cross-lingual generation.

B. MULTILINGUAL TRANSLATION

In contrast to traditional Neural Machine Translation (NMT), which handles text-to-text conversion, Text-to-Speech Translation (T2ST) offers several practical advantages, such as facilitating the development of cross-lingual audiobooks and broadcasting systems. The first such approach was introduced by Mingote et al. [1], who extended mBART for direct text-to-speech translation by incorporating discrete speech units. Their pipeline discretizes target-language speech with a speech tokenizer and then fine-tunes mBART to predict speech units from source text, applying U2S to synthesize the speech [14], [18].

However, languages differ significantly in vocabulary, syntax, and pronunciation, and their speech outputs exhibit diverse prosodic nuances in such forms as intonation and rhythm. Consequently, models without task- or language-specific specialization often yield suboptimal performance. Additionally, the standard mBART architecture, which uses a shared Transformer backbone, may not be optimal for handling the diversity of languages.

To overcome these challenges, we propose a novel model called SAM-Translator, which incorporates two key innovations: SPL [5], [6] and MoE [7], [8]. SPL enhances the learning process by progressively selecting training examples

based on increasing difficulty, as determined by the model's own loss. This approach improves learning efficiency across diverse languages. The MoE mechanism uses multiple expert subnetworks to dynamically route inputs to the most relevant experts, enabling the model to provide language- or task-specific representations.

Experimental results demonstrate that SAM-Translator achieves significant improvements on the standard benchmarks, i.e., the CVSS-C and CoVoST-2 datasets.

C. LOW-RESOURCE PERFORMANCE

Despite significant progress in multilingual translation, the performance of T2ST models remains limited for low-resource languages due to the scarcity of training data. To mitigate this, we implement DAST, a semi-supervised model widely used in NMT.

DAST first trains a baseline model on the available parallel corpus and then uses it to generate pseudo-translations for monolingual sentences. The resulting synthetic corpus, combined with the original data, has been shown to significantly improve translation quality, particularly when the synthetic data are carefully curated [19], [20].

In this work, we extend the DAST approach to the domain of T2ST to alleviate the data scarcity for low-resource languages. Experimental results demonstrate that this method significantly enhances translation performance, validating the feasibility and effectiveness of DAST in data-constrained scenarios for T2ST.

III. PROPOSED FRAMEWORK

A. OVERALL PIPELINE

To perform T2ST, we formulate the problem as follows (see Figure 1).

Step 1: Extracting Speech Units. Discrete speech units ($\hat{U}_t$) are extracted from target speech S_t using the HuBERT-Base+KM200 model.

$$\hat{U}_t = \text{HuBERT-Base+KM200}(S_t). \quad (1)$$

Step 2: T2U Transformation. In the second stage, the source text (T_s) is transformed into a sequence of target units ($\tilde{U}_t$) through a T2U model.

$$\tilde{U}_t = \text{T2U}(T_s). \quad (2)$$

Step 3: U2S Synthesis. Finally, a speech synthesis model is used to transform the generated unit sequence ($\tilde{U}_t$) into a speech signal ($\hat{S}_t$).

$$\hat{S}_t = \text{U2S}(\tilde{U}_t). \quad (3)$$

B. SPEECH UNIT EXTRACTION

In this study, we adopt a pretrained self-supervised HuBERT model [2] to extract speech representations from the 8th layer. These representations are discretized via K-means clustering with $K = 200$ centroids. Each discrete unit represents a 20-ms segment of the 16-kHz raw audio (S_t), corresponding to 50 units per second (1000 ms). However, the

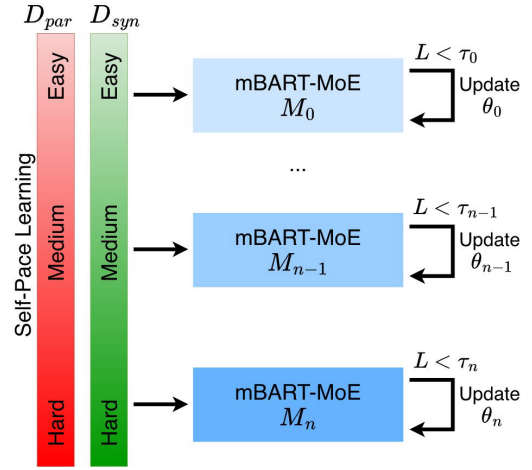


FIGURE 2. Proposed SAM-Translator Framework: Incorporating SPL and MoE mechanisms to generate accurate speech units across diverse languages.

discrete unit sequence often contains consecutive repetitions of the same unit. To address this issue, we apply a lossy encoding method [21] to collapse them (e.g., 1, 1, 1, 2, 2 $\rightarrow$ 1, 2), producing a compact semantic sequence. Next, this sequence is converted to synthesized speech using the Speech Decoder (detailed in Section IV-C). Here, the decoder implicitly recovers the duration information discarded during the encoding phase, thus restoring natural prosody in the synthesized speech.

C. SELF-PACED LEARNING

In our proposed framework, we incorporate SPL [5], [6], a curriculum learning strategy that emulates human learning by prioritizing simpler samples before addressing more complex ones. By assessing sample difficulty through the model's loss, our mBART-MoE model dynamically selects training samples in each iteration. Specifically, samples with a per-sample loss averaged over the sequence length, below a threshold τ are considered "easy" and included in training, while the others are excluded. This selection is formalized by a binary weight function:

$$v_i = \begin{cases} 1 & \text{if } l_{CE}(\mathbf{w}, t_i, u_i) < \tau, \\ 0 & \text{otherwise,} \end{cases} \quad (4)$$

where $l_{CE}(\mathbf{w}, t_i, u_i)$ denotes the cross-entropy loss of the model with parameters $\mathbf{w}$ on sample (t_i, u_i) , and $v_i \in \{0, 1\}$ indicates whether the sample contributes to the training objective. The training loss is computed as a weighted mean over selected samples:

$$\mathcal{L}_{CE} = \frac{\sum_{i=1}^N v_i l_{CE}(\mathbf{w}, t_i, u_i)}{\sum_{i=1}^N v_i}. \quad (5)$$

As training progresses (Figure 2), τ is incrementally increased until it reaches the point where 100% of the training samples are included. This incremental strategy enables the model to build its learning capacity in each stage, improving

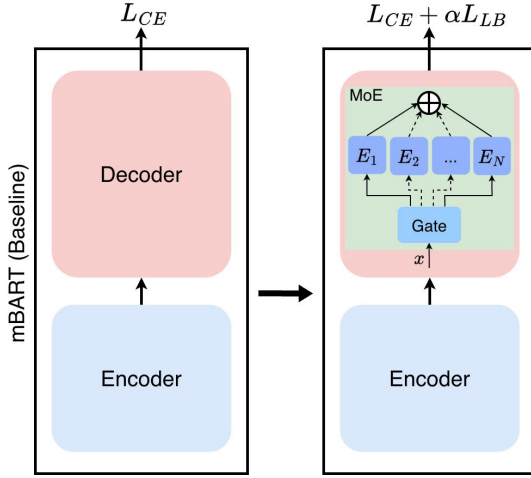


FIGURE 3. Proposed mBART-MoE Model: Integrating mBART with MoE to enhance language adaptability.

stability and robustness. All hyperparameters are configured as detailed in IV-C.

D. SPARSE MIXTURE OF EXPERTS

In this study, the Feedforward Network (FFN) of the baseline mBART [3] model is replaced with the MoE [7], [8] module to achieve greater capacity at similar compute costs due to sparsity, as illustrated in Figure 3. The MoE architecture incorporates N expert networks, which are dynamically selected via a gating mechanism. The output of the MoE layer is computed as

$$\text{Out} = \sum_{i=0}^{N-1} G_i(x) \cdot E_i(x), \quad (6)$$

where $x \in \mathbb{R}^{B \times S \times H}$ represents the input tensor (B : batch size, S : sequence length, H : hidden size), $G_i(x)$ denotes the gating weight for the i -th expert, and $E_i(x)$ is the corresponding expert's output. The gating weights are determined by

$$G(x) = \text{Softmax}(\text{TopK}(x \cdot W_g)), \quad (7)$$

where $W_g \in \mathbb{R}^{H \times N}$ are learnable gating parameters and $\text{TopK} = 2$ for token-level routing. This setup ensures high capacity and reduced computational overhead by processing all tokens with their assigned Top- K experts.

Each expert is implemented as follows:

$$E_i(x) = [\text{ReLU}(x \cdot W_{1,i}) \odot (x \cdot W_{2,i})] \cdot W_{3,i} \quad (8)$$

where $\odot$ denotes element-wise multiplication, $W_{1,i}, W_{2,i} \in \mathbb{R}^{H \times I}$ are projection matrices with an intermediate size of $I = 4H$, and $W_{3,i} \in \mathbb{R}^{I \times H}$ maps the intermediate representation back to the hidden dimension.

To ensure balanced utilization of experts and prevent over-reliance on a subset of them, a load balancing loss $\mathcal{L}_{LB}$ is incorporated in the training objective. This loss penalizes uneven token assignments across experts and is formulated

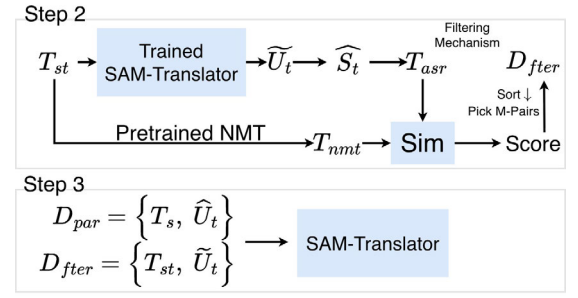


FIGURE 4. Process of generating synthetic data to improve the performance on low-resource languages.

as:

$$\mathcal{L}_{LB} = \alpha \cdot N \cdot \sum_{i=0}^{N-1} f_i \cdot P_i, \quad (9)$$

where

$$f_i = \frac{1}{B \cdot S} \sum_{b,s} \mathbb{I}\{i \in \text{TopK}(x \cdot W_g)\}, \quad (10)$$

measures the fraction of tokens routed to expert i

$$P_i = \frac{1}{B \cdot S} \sum_{b,s} G_i(x) \quad (11)$$

represents the average gating probability for expert i , and α is a weighting coefficient.

E. DATA-AUGMENTED SELF-TRAINING

Given a parallel text-speech corpus $\mathcal{D}_{par} = \{T_s^n, \hat{U}_t^n\}_{n=1}^N$, T_s^n is a source text sentence in a given language and $\hat{U}_t^n$ is the corresponding sequence of discrete acoustic units derived from the target speech. In addition, we have a large-scale multilingual text-only corpus $\mathcal{D}_{st} = \{T_{st}^m\}_{m=1}^M$ without paired speech. The goal is to leverage $\mathcal{D}_{st}$ to augment $\mathcal{D}_{par}$ and improve the T2ST model's performance. To achieve this, we use a DAST method carried out in three steps, as shown in Figure 4:

- **Step 1 – Initial Training:** Train the SAM-Translator model on the parallel corpus $\mathcal{D}_{par}$.
- **Step 2 – Synthetic Pair Generation & Filtering:** Use the trained SAM-Translator to generate discrete unit sequences $\tilde{U}_t^m$ for each source text T_{st}^m in $\mathcal{D}_{st}$.

$$\tilde{U}_t^m = \text{SAM-Translator}(T_{st}^m),$$

thus forming synthetic dataset $\mathcal{D}_{syn} = \{(T_{st}^m, \tilde{U}_t^m)\}_{m=1}^M$. A filtering process (Algorithm 1) is then applied to obtain a high-quality subset ($\mathcal{D}_{fiter}$).

- **Step 3 – Retraining:** Retrain the SAM-Translator on the combined corpus $\mathcal{D}_{par} \cup \mathcal{D}_{fiter}$.

Filtering Mechanism: To remove low-quality synthetic pairs, we apply the filtering mechanism described in Algorithm 1 (Step 2 in Figure 4). Specifically, for each synthetic pair $(T_{st}^m, \tilde{U}_t^m)$, the acoustic unit sequence $\tilde{U}_t^m$

IDX Persian (Highest Bleu Score on Low Resource)	
0	Text input: مردی در حضور مردی دیگر در شب سیگار می کشد.
	SAM-Translator (ASR): one man smokes in the presence of another man at night
	NMT: one man smokes in the presence of another man at night
	Similar Score: 1
9999	Text input: یک شمارشگر "جونز" به دوچرخه متصل بود تا طول مسیر را اندازه گیری کند.
	SAM-Translator (ASR): the johns meter was connected to the bike to measure the whole route
	NMT: a jones counter was attached to the bike to measure the length of the ride
	Similar Score: 0.74
Tamil (Lowest Bleu Score on Low Resource)	
0	Text input: அது என்ன என்பதை நான் தெரிந்துகொள்ள விரும்புகிறேன்
	SAM-Translator (ASR): I want to know what it is
	NMT: I want to know what it is
	Similar Score: 1
9999	Text input: வீட்டிலிருந்து பணியாற்றுவது குறைபாடுகள் மற்றும் நன்மைகள் என இரண்டையும் கொண்டிருக்கிறது.
	SAM-Translator (ASR): working from the house has both disadvantages and benefit
	NMT: working from home has both disadvantages and benefits
	Similar Score: 0.70

FIGURE 5. Synthetic data examples from DAST method.

is converted into synthesized speech $\hat{S}_t$ using the Speech-Decoder as described in IV-C. A pretrained ASR then transcribes $\hat{S}_t$ into text T_{asr} . Furthermore, a pretrained NMT model translates the same source text T_{st}^m into the target text T_{nmt} .

Next, we measure the textual consistency between T_{asr} and T_{nmt} using the cosine similarity of their TF-IDF embeddings. The synthetic pairs are ranked according to this similarity score, and the top-M pairs with the highest scores are selected to form the filtered dataset $\mathcal{D}_{fiter}$. Languages not supported by the NMT model (pretrained mBART [3]) (e.g., Welsh) are excluded from this process. Additionally, for all languages, we apply a filtering procedure to ensure that only well-formed synthetic samples are used during the retraining in Step 3. Several qualitative examples are presented in Figure 5. Further details of the experiment setup are given in Section IV-B.

Algorithm 1 Filtering Mechanism

Input: Synthetic dataset $\mathcal{D}_{syn} = \{T_{st}^m, \tilde{U}_t^m\}_{m=1}^M$
Output: Filtered dataset $\mathcal{D}_{fiter}$
foreach $(T_{st}^m, \tilde{U}_t^m) \in \mathcal{D}_{syn}$ **do**
 $\hat{S}_t \leftarrow \text{Speech-Decoder}(\tilde{U}_t^m)$
 $T_{asr} \leftarrow \text{ASR}(\hat{S}_t)$
 $T_{nmt} \leftarrow \text{NMT}(T_{st}^m)$
 $\text{Score}_m \leftarrow \text{SIM}(\text{TF-IDF}(T_{asr}), \text{TF-IDF}(T_{nmt}))$
end
Rank all pairs by Score_m in descending order
Select top-M pairs $\Rightarrow \mathcal{D}_{fiter}$
return $\mathcal{D}_{fiter}$

IV. EXPERIMENT SETUP

A. DATASET

This study uses two datasets. The first is the CVSS-C dataset [22], a large multilingual corpus comprising text from 21 languages paired with corresponding English speech. The speech samples in CVSS-C are synthesized by a

single-speaker TTS model but have high naturalness and clarity. The details of the 21 language sources and their respective data sizes are provided in Appendix B, Table 3.

The second dataset is CoVoST-2, which consists of speech translations from 15 languages into English. This large-scale multilingual corpus uses English speech from Common Voice version 4, recorded using a variety of devices and speakers. Although CoVoST-2 is primarily designed for Speech-to-Text (S2T) tasks, we use it here for the T2ST purposes. Specifically, we align the provided source text translations with their corresponding ground-truth English speech and then remove samples with empty text. The final data sizes for the training, validation, and test splits are approximately 209K, 14K, and 14K, respectively.

B. EXPERIMENT SETUP

For the CVSS dataset, the baseline mBART model is trained first, followed by the incorporation of SPL, and finally, the incorporation of MoE (SAM-Translator). Next, for the DAST approach, instead of augmenting data for all languages, we focus on those with low BLEU scores. Specifically, we select languages with BLEU scores below 20, including Chinese, Japanese, Mongolian, Persian, Tamil, and Welsh. These languages are also low-resource languages, with real parallel data sizes smaller than 10K samples. In this experiment, we use M-pairs of 5K, 10K, and 15K filtered synthetic samples per language to constitute the augmented dataset ($\mathcal{D}_{filter}$). We limit the synthetic data size because we have observed no significant improvement with larger datasets. The source-side text corpus for these languages is derived from CoVoST-2 (X-into-English) [23], which is large but contains many repeated samples. To ensure diversity during learning, we use only unique text for synthesizing and filtering, enabling the model to learn from a broader range of contexts rather than from a set of repetitive samples.

For the CoVoST-2 dataset, we follow a similar approach as that taken with CVSS. Specifically, we first use SPL in the baseline mBART model, followed by the integration of MoE. Note that, since this is a high-resource dataset, the DAST approach is not applied.

C. IMPLEMENTATION PARAMETERS

In this research, we use mBART as the baseline model for T2ST, initializing its parameters with a pretrained mBART-50 (many-to-one) model. The same pretrained model is also used to initialize the proposed SAM-Translator framework. Both the baseline and SAM-Translator consist of 12 encoder layers and 12 decoder layers, each with 16 attention heads.

We train the models using the AdamW optimizer [24] with a learning rate of 3×10^{-5} , $\beta_1 = 0.9$, $\beta_2 = 0.999$ and a batch size of 32. The baseline model has 611M parameters, while mBART-MoE employs eight experts with Top-K ($K = 2$) with 1.7B parameters. Additionally, a load-balancing loss with a coefficient $\alpha = 0.001$ is incorporated in the primary cross-entropy loss. For the SPL hyperparameters, we start with

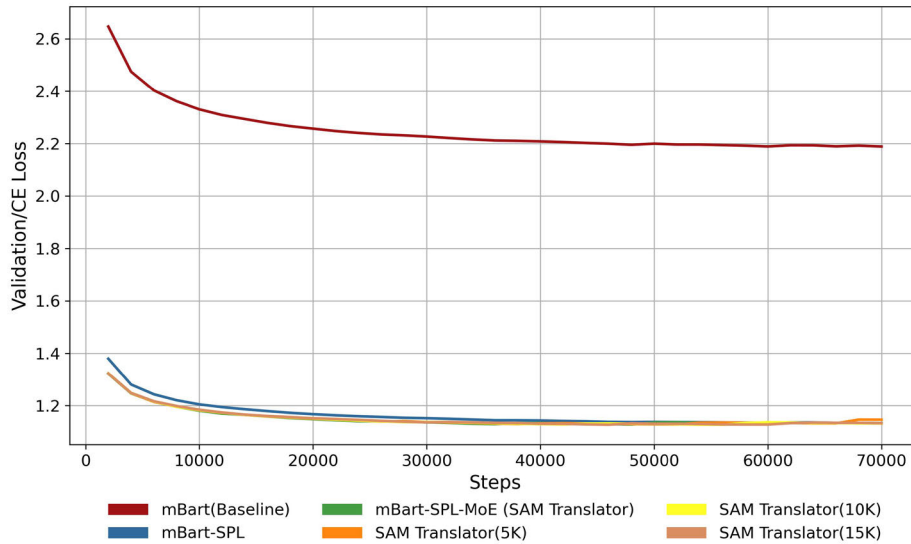


FIGURE 6. Validation cross-entropy loss of the baseline (mBART) and proposed SAM-Translator models. After SAM-Translator, 5K, 10K, 15K indicate synthetic sample sizes for self-training. (Best viewed in color).

$\tau = 2$ and increase it by 0.25 every 2000 steps until 100% of the training data samples are included.

During training, model checkpoints are saved every 2,000 steps, and the final evaluation is based on the average of the last five checkpoints. All experiments are conducted on an NVIDIA A100 GPU with 80 GB of memory. The SAM-Translator model takes approximately 1.5–2 days to train; however, training automatically stops if the validation loss does not decrease over five consecutive evaluations.

To convert unit sequences into speech, we use the Speech-Decoder, implemented with Grad-TTS [25], an end-to-end speech synthesis model that generates high-quality, natural audio.

D. EVALUATION METRICS

To evaluate T2ST, we use a pretrained automatic speech recognition (ASR) model¹ to transcribe the generated speech into text. The resulting transcriptions are then compared to the reference translations using the BLEU score [26], providing a quantitative measure of outputs' translation accuracy.

For the cascade setup, we first use a pretrained mBART model to translate the source text into the target text. The translated target text is then synthesized into speech with a TTS model [15], and this speech is subsequently transcribed back into text using the same ASR model employed in T2ST.

In addition, to assess the speech quality of the Speech Decoder, we compared two models, speech unit to speech and text to speech, using two metrics: naturalness and intelligibility. In this experiment, 10 participants evaluated 15 samples randomly presented for each model (Speech-Unit, Text-to-Speech, and Ground-Truth), for a total of 45 samples

per participant. The samples were rated on a 5-point scale for naturalness and intelligibility (see Appendix A).

TABLE 1. Subjective evaluation: Speech-Unit and Text-Based speech synthesis with 95% confidence intervals.

Method	Naturalness	Intelligibility
Ground Truth	4.35 ± 0.13	4.53 ± 0.10
Text to Speech	4.15 ± 0.14	4.36 ± 0.12
Speech-Unit to Speech (Speech Decoder)	4.12 ± 0.14	4.24 ± 0.12

V. RESULTS AND DISCUSSION

A. SPEECH QUALITY

In this experiment, as shown in Table 1, we evaluated the speech quality of two models: the speech unit (SAM-Translator's speech decoder) and the TTS model (pretrained Grad-TTS). The naturalness of both the input text and the speech unit is comparable, with ratings of 4.15 ± 0.14 and 4.12 ± 0.14 , respectively. However, the intelligibility of the input text is slightly higher than that of the speech unit, with scores of 4.36 ± 0.12 and 4.24 ± 0.12 , respectively. This difference is due to the characteristic of text to explicitly convey semantic information. Nevertheless, the intelligibility of the speech unit, at 4.24 ± 0.12 , is still quite high, making it well-suited for applications that do not rely on text or in languages without written transcription.

B. CVSS DATASET

1) EFFECTIVENESS OF SAM-TRANSLATOR

The effectiveness of the SPL mechanism is evident in its impact on validation loss, which it significantly reduced

¹jonatasgrosman/wav2vec2-large-xlsr-53-english

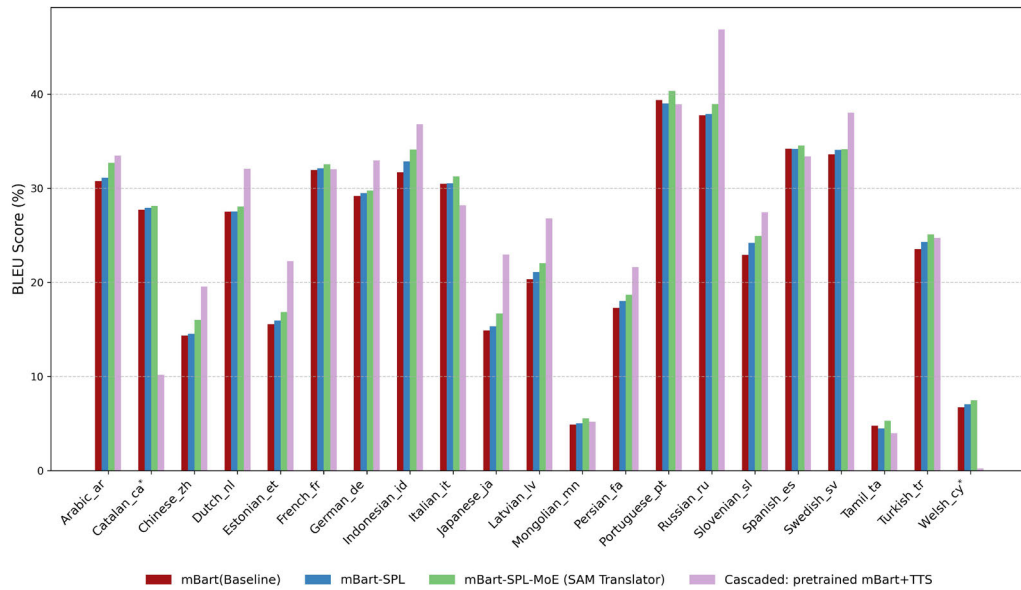


FIGURE 7. BLEU scores on CVSS-C for three methods; * marks languages not in pretrained mBART-50 (NMT). (Best viewed in color).

TABLE 2. BLEU scores and relative improvements of SAM Translator with data-augmented self-training. Here, 5K, 10K, and 15K indicate the synthetic sample sizes after the filtering process.

Model	mBart (Baseline)	SAM-Translator	SAM-Translator (5K)	SAM-Translator (10K)	SAM-Translator (15K)
Chinese_zh	14.36	16.01 (+11.52%)	16.23 (+13.02%)	16.41 (+14.31%)	16.56 (+15.35%)
Japanese_ja	14.91	16.68 (+11.86%)	17.16 (+15.06%)	17.68 (+18.58%)	17.38 (+16.53%)
Mongolian_mn	4.91	5.55 (+13.14%)	6.09 (+24.05%)	6.18 (+25.83%)	5.99 (+22.02%)
Persian_fa	17.29	18.69 (+8.12%)	19.61 (+13.42%)	20.04 (+15.92%)	20.03 (+15.90%)
Tamil_ta	4.77	5.29 (+10.75%)	5.38 (+12.83%)	5.50 (+15.26%)	5.68 (+19.01%)
Welsh_cy*	6.73	7.46 (+10.85%)	8.18 (+21.55%)	8.19 (+21.69%)	7.63 (+13.30%)

during training (Figure 6). Additionally, SPL improved the baseline mBART model in 18 out of 21 languages, yielding an overall relative BLEU improvement of approximately 1–2%.

The results in Figure 7 demonstrate that SAM-Translator consistently outperforms the baseline mBART across all evaluated languages. Compared with the cascaded model, SAM-Translator shows substantially better performance on languages such as Catalan and Welsh, likely because the pretrained mBART model was not trained on these languages. Conversely, for languages such as Russian, the cascaded model performs better than SAM-Translator. This difference is attributed to the scale of training data: the pretrained mBART model was trained on approximately 14 million text pairs, whereas SAM-Translator was trained on 10K pairs. Moreover, mBART operates at the text level, which conveys clear semantic meaning, whereas SAM-Translator works with speech units, which are less explicit. Although SAM-Translator gives slightly lower performance than the

cascaded model at a similar inference speed, it still offers distinct practical advantages in real-world applications.

To further analyze performance, we divided the examined languages into two groups based on their training data size: high-resource (more than 10,000 training samples) and low-resource (fewer than 10,000 training samples). The high-resource group includes French, German, Spanish, Catalan, Italian, and Russian. For these languages, the SAM-Translator model demonstrated modest improvements, with BLEU score gains ranging from approximately 2% to 3%. In contrast, low-resource languages exhibited more significant improvements. Notably, languages such as Chinese, Indonesian, Japanese, and Swedish achieved significant relative gains of around 7–10%. For instance, despite having only approximately 1,200 training samples, Indonesian achieved an impressive BLEU score of 34.96. Similarly, Tamil and Mongolian showed substantial relative improvements of 10% and 13%, respectively, although their BLEU scores remained

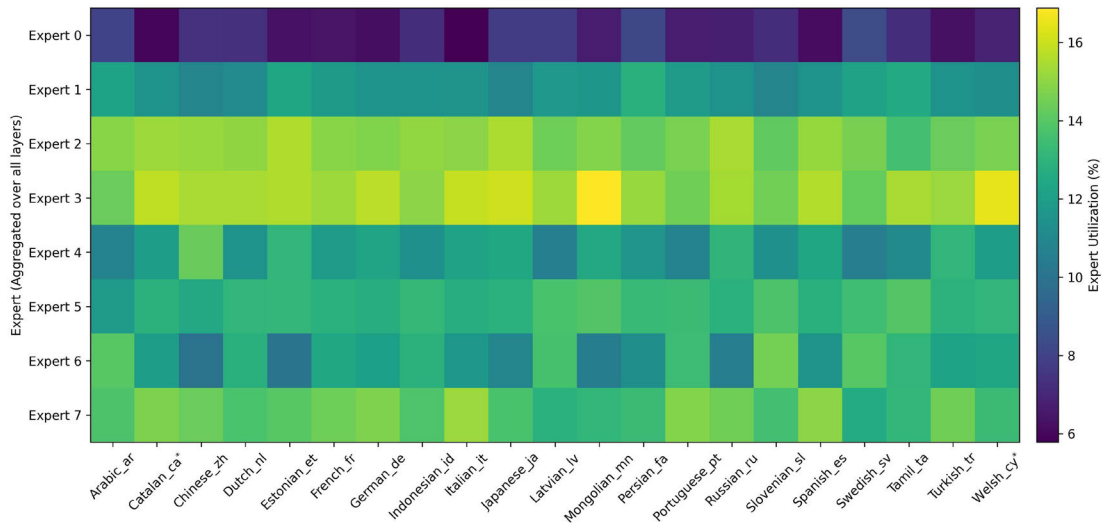


FIGURE 8. Heatmap of expert selection distribution across languages of SAM-Translator. * marks languages not in pretrained mBART-50 (NMT). (Best viewed in color).

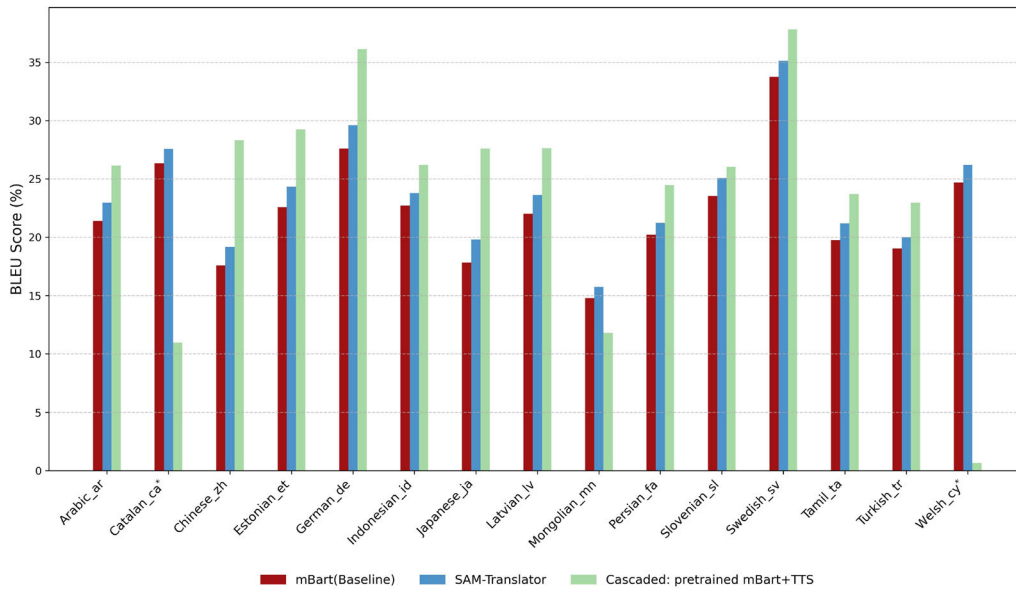


FIGURE 9. BLEU scores on CoVoST-2; * marks languages not in pretrained mBART-50 (NMT). (Best viewed in color).

relatively low at 5.38 and 5.55, reflecting the inherent challenges of working with limited training data.

An interesting trend could be observed from the performances of languages not included in the pretraining of the original mBart model. For example, Catalan, which is absent from the pretrained mBart-50 (NMT), achieved a high BLEU score of 28.39, representing a relative improvement of 1.49%. Similarly, Welsh, another language not present in the pretraining data, showed a significant relative improvement of 10.85%, despite having a lower BLEU score of 7.46. These results suggest that both languages can benefit from their shared linguistic family (Indo-European group) with

high-resource languages like French and Spanish. However, the difference in their performances highlights the importance of training data size, as seen by Catalan, with its larger training dataset, outperforming Welsh.

2) EFFECTIVENESS OF DATA-AUGMENTED SELF-TRAINING

The DAST approach markedly improves performance for low-resource languages, as shown in Table 2. Specifically, generating 10K synthetic samples through self-training yields substantial BLEU score gains across all target languages, with Welsh and Mongolian achieving relative improvements of 21.69% and 25.83%, respectively. However, increasing the

CVSS Portuguese (Highest Bleu Score) Text input: o juiz clark achou difficil resolver esse desacordo Correct: SAM-Translator: judge clark found it difficult to solve this disagreement Ground Truth: the judge clark found it hard to resolve this disagreement Text input: uma senhora de branco senta-se num tapete com uma bola na mão. Incorrect: SAM-Translator: a white lady sitting on a carpet with a ball in her hand Ground Truth: a lady in white sits on a rug with a ball in her hand Tamil (Lowest Bleu Score - Low Resource) Text input: ஆதிமுதல் இந்தத் தேதி வரைக்கும் Correct: SAM-Translator: to this date from the first time Ground Truth: from the beginning till date Text input: துணைக்கோள்களே இல்லாத கோள்கள் புதன் மற்றும் வெள்ளி Incorrect: SAM-Translator: the classes without satellites are boodin silver Ground Truth: mercury and venus planets without satellites	CoVoST-2 Swedish (Highest Bleu Score) Text input: den långlivade bron står fortfarande på sin plats i dag Correct: SAM-Translator: the long life bridge still stands in its place today Ground Truth: the long-lived bridge still stands today Text input: hans tal avslutades med en stående ovation Incorrect: SAM-Translator: his speech ended with a standing over Nile Ground Truth: his speech ended with a standing ovation Mongolian (Lowest Bleu Score) Text input: энэ бол европын театруудын холбооны гишүүн театр мөн. Correct: SAM-Translator: it is also a member of the federation of european theatres Ground Truth: it is a member theater of the union of the theatres of europe Text input: суралчид өнгийн харандаагаараа будав. Incorrect: SAM-Translator: students painted with colors of colors Ground Truth: the students colored with their crayons.
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FIGURE 10. Qualitative evaluation of SAM-Translator on CVSS and CoVoST-2 datasets. Common errors, such as repetition and inaccuracy in semantic representation, are highlighted in red.

synthetic data to 15K samples does not consistently further enhance performance. For languages such as Japanese, Mongolian, and Welsh, performance even declines, suggesting that excessive synthetic data may introduce noise or domain shift, potentially reducing the model's effectiveness.

Illustrative examples of Welsh translations are presented below for SAM-Translator trained with 10K and 15K synthetic samples.

- Input: bydd o'n dechrau am wyth o'r gloch
- Target: It will start at eight o'clock
- SAM-Translator(10K): It will start at eight o'clock
- SAM-Translator(15K): It will start and aviate the sound

Clearly, SAM-Translator (10K) provides a more accurate translation, while SAM-Translator (15K) introduces some errors.

Additionally, all remaining languages not augmented with synthetic data show minor BLEU variations of approximately 1–2%. For detailed results, see Appendix B, Table 3.

3) ROUTING VISUALIZATION

Figure 8 illustrates the distribution of selected experts across all languages during inference. The results show differences in expert selection for each language, indicating that the model dynamically triggers specialized experts to tailor the linguistic and acoustic characteristics of each language. This visualization underscores the effectiveness of our mBart-MoE model in optimizing performance across diverse linguistic contexts.

C. COVOST-2 DATASET

SAM-Translator consistently outperforms mBart (Baseline) across all languages in the CoVoST-2 dataset, as presented in Figure 9. The average relative improvement ranges from 5% to 7%, with a particularly notable increase of 11.2% for Japanese. Among the languages in the CoVoST-2 dataset, Swedish achieves the highest performance, with a BLEU score improvement of 35.12%, while Mongolian exhibits the lowest performance, with a BLEU score of 15.74%. Although the cascaded method demonstrates superior performance to that of SAM-Translator, it must be emphasized that SAM-Translator does not rely on text as an intermediate representation. Detailed results are given in Table 4.

D. QUALITATIVE RESULTS

Figure 10 presents several examples of SAM-Translator's performance on two datasets: CVSS and CoVoST-2. In this qualitative evaluation, we present representative examples for each language, selected based on specific criteria such as the languages with the highest and lowest BLEU scores. For correct examples, the quality of SAM-Translator's output closely matches the ground truth. In contrast, the incorrect examples exhibit various errors (highlighted in red), including inaccurate semantic representation, inappropriate word choice, and word repetition.

VI. CONCLUSION

In this work, we proposed SAM-Translator, a novel framework for T2ST that incorporates SPL and MoE mechanisms in the mBART architecture. SAM-Translator effectively enhances multilingual speech translation by progressively learning from easy to hard samples and dynamically routing inputs to specialized subnetworks, thus improving adaptation to linguistic diversity. Additionally, we use the DAST technique to generate synthetic training data, addressing the challenges posed by low-resource languages. Experimental evaluations on the CVSS-C dataset demonstrate that SAM-Translator outperforms baseline methods, with significant improvements with both high- and low-resource languages. Moreover, this model exhibits strong generalization capability and robustness on the CoVoST2 dataset. However, SAM-Translator yields relatively low scores for low-resource languages such as Welsh and Tamil, and its outputs occasionally contain such errors as inaccurate semantic representation and word repetition. For future work, we aim to further improve performance with low-resource languages, mitigate the current limitations, and extend our approach to additional source and target languages.

APPENDIX A SUBJECTIVE METRIC

Intelligibility Rating Scale:

- 1) Unintelligible: Speech is entirely unclear.
- 2) Poor: Only a few words are recognizable.
- 3) Fair: Some segments are intelligible, but many errors remain.

TABLE 3. Performance comparison of different models across languages on CVSS-C. Relative improvements over baseline mBART are shown as percentage gains next to the corresponding scores.

	Data Size			mBART	mBART-SPL	SAM-Translator	SAM-Translator (5K)	SAM-Translator (10K)	SAM-Translator (15K)	Cascaded
	train	dev	test							
Arabic_ar	2029	1758	1693	30.76	31.14 (+1.20%)	32.70 (+6.29%)	33.45 (+8.74%)	34.31 (+11.51%)	33.09 (+7.55%)	33.46
Catalan_ca*	54171	12730	12730	27.73	27.92 (+0.69%)	28.14 (+1.49%)	28.39 (+2.39%)	28.33 (+2.18%)	28.09 (+1.30%)	10.17
Chinese_zh	7079	4843	4897	14.36	14.55 (+1.34%)	16.01 (+11.52%)	16.23 (+13.02%)	16.41 (+14.31%)	16.56 (+15.35%)	19.56
Dutch_nl	1893	1699	1699	27.51	27.54 (+0.09%)	28.07 (+2.03%)	29.58 (+7.51%)	28.68 (+4.23%)	28.61 (+3.98%)	32.07
Estonian_et	1750	1576	1571	15.57	15.96 (+2.46%)	16.85 (+8.20%)	17.21 (+10.51%)	17.87 (+14.75%)	17.51 (+12.44%)	22.27
French_fr	130597	14759	14759	31.94	32.13 (+0.61%)	32.56 (+1.94%)	32.43 (+1.54%)	32.56 (+1.94%)	32.54 (+1.90%)	32.04
German_de	71830	13511	13504	29.19	29.51 (+1.06%)	29.77 (+1.99%)	30.21 (+3.49%)	30.35 (+3.98%)	29.89 (+2.39%)	32.98
Indonesian_id	1243	792	844	31.70	32.88 (+3.58%)	34.13 (+7.67%)	34.96 (+10.26%)	35.42 (+11.74%)	35.51 (+12.02%)	36.81
Italian_it	19387	8938	8951	30.49	30.53 (+0.13%)	31.26 (+2.54%)	31.62 (+3.71%)	31.55 (+3.48%)	31.22 (+2.39%)	28.19
Japanese_ja	1119	634	684	14.91	15.33 (+2.74%)	16.68 (+11.86%)	17.16 (+15.06%)	17.68 (+18.58%)	17.38 (+16.53%)	22.96
Latvian_lv	2336	1125	1629	20.35	21.09 (+3.50%)	22.05 (+8.34%)	22.96 (+12.82%)	23.31 (+14.51%)	22.36 (+9.83%)	26.80
Mongolian_mn	2018	1760	1759	4.91	5.03 (+2.39%)	5.55 (+13.14%)	6.09 (+24.05%)	6.18 (+25.83%)	5.99 (+22.02%)	5.20
Persian_fa	4346	3440	3425	17.29	18.02 (+4.08%)	18.69 (+8.12%)	19.61 (+13.42%)	20.04 (+15.92%)	20.03 (+15.90%)	21.63
Portuguese_pt	6014	3318	4023	39.37	39.01 (-0.92%)	40.34 (+2.47%)	40.52 (+2.92%)	40.24 (+2.20%)	40.80 (+3.64%)	38.94
Russian_ru	10352	6110	6300	37.75	37.90 (+0.38%)	38.95 (+3.17%)	39.49 (+4.61%)	39.40 (+4.38%)	39.08 (+3.51%)	46.88
Slovenian_sl	1566	509	360	22.92	24.20 (+5.28%)	24.94 (+8.82%)	25.39 (+10.78%)	25.84 (+12.74%)	24.82 (+8.28%)	27.46
Spanish_es	64350	13212	13216	34.21	34.17 (-0.12%)	34.56 (+1.02%)	34.91 (+2.04%)	34.67 (+1.33%)	34.64 (+1.26%)	33.40
Swedish_sv	2152	1349	1595	33.59	34.08 (+1.44%)	34.15 (+1.64%)	35.56 (+5.86%)	36.01 (+7.19%)	35.45 (+5.54%)	38.03
Tamil_ta	1358	384	786	4.77	4.49 (-6.22%)	5.29 (+10.75%)	5.38 (+12.83%)	5.50 (+15.26%)	5.68 (+19.01%)	3.99
Turkish_tr	1823	1623	1629	23.56	24.30 (+3.08%)	25.08 (+6.49%)	26.55 (+12.71%)	26.54 (+12.66%)	26.42 (+12.15%)	24.73
Welsh_cy*	721	688	690	6.73	7.06 (+4.55%)	7.46 (+10.85%)	8.18 (+21.55%)	8.19 (+21.69%)	7.63 (+13.30%)	0.23

Boldfaced scores are statistically significant ($p < 0.01$) in the paired t-test comparing SAM-Translator to mBART.

All experiments are evaluated using Wav2Vec 2.0. Preliminary evaluations with Whisper showed a similar trend, with the SAM-Translator outperforming the baseline, so we focus on Wav2Vec 2.0 results in this paper.

TABLE 4. Performance comparison of different models across languages on CoVoST-2. Relative improvements over baseline mBart are shown as percentage gains next to the corresponding scores.

Language	mBart	SAM Translator	Cascaded	Language	mBart	SAM Translator	Cascaded
Arabic_ar	21.40	22.97 (+7.33%)	26.15	Mongolian_mn	14.77	15.74 (+6.59%)	11.81
Catalan_ca*	26.32	27.56 (+4.71%)	10.95	Persian_fa	20.21	21.22 (+4.98%)	24.47
Chinese_zh	17.60	19.17 (+8.95%)	28.29	Slovenian_sl	23.53	25.05 (+6.46%)	26.03
Estonian_et	22.58	24.32 (+7.68%)	29.23	Swedish_sv	33.74	35.12 (+4.10%)	37.80
German_de	27.59	29.59 (+7.25%)	36.12	Tamil_ta	19.74	21.20 (+7.35%)	23.71
Indonesian_id	22.72	23.80 (+4.75%)	26.20	Turkish_tr	19.03	20.01 (+5.17%)	22.97
Japanese_ja	17.81	19.80 (+11.20%)	26.59	Welsh_cy*	24.68	26.20 (+6.16%)	0.64
Latvian_lv	22.01	23.62 (+7.32%)	27.63	—	—	—	—

Boldfaced scores are statistically significant ($p < 0.01$) in the paired t-test comparing SAM-Translator to mBART.

- 4) Good: Largely intelligible with minor artifacts.
- 5) Perfect: Fully intelligible, requiring no effort to understand meaning.

Naturalness Rating Scale:

- 1) Very Unnatural: Speech sounds robotic or synthetic.
- 2) Unnatural: Noticeable artificiality in speech.
- 3) Neutral: Neither clearly natural nor clearly unnatural.
- 4) Natural: Speech sounds mostly natural with minor artifacts.
- 5) Very Natural: Fully natural, indistinguishable from human speech.

APPENDIX B COMPREHENSIVE BLEU SCORES

See Tables 3 and 4.

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